

# 114-3□□□□□\_Circular Motion\_□□

## Q1

Imagine a flat playground spinner. The distance between the center and the edge of the spinner is  $r = 3\text{ m}$ . It is a perfectly flat, solid circular platform that spins horizontally.

- (a) The spinner rotates through an angle of  $\theta = 4\text{ rad}$ . Calculate the actual distance you traveled if you sit at the edge.
- (b) If it takes  $t = 2\text{ s}$  to complete this rotation, calculate the angular velocity ( $\omega$ ) of the spinner.
- (c) Calculate your linear speed.



## Q2

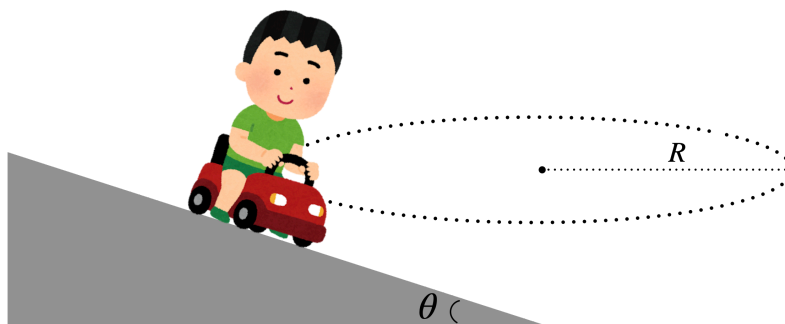
Imagine you place a small coin on the very edge of the same spinner from Q1.

- (a) Calculate the centripetal acceleration of the coin while it is turning on the spinner.
- (b) Which direction does this centripetal acceleration point? Depict circle as the top view of the spinner, and denote the acceleration arrow.
- (c) Suddenly, the coin slips! Which direction does the coin fly in the horizontal direction? Explain using Newton's 1st Law.

### Q3

Imagine a 1200 kg car turning on a smooth, frictionless road that is banked (tilted) at an angle of  $\theta = 37^\circ$ . The car is making a HORIZONTAL circle.

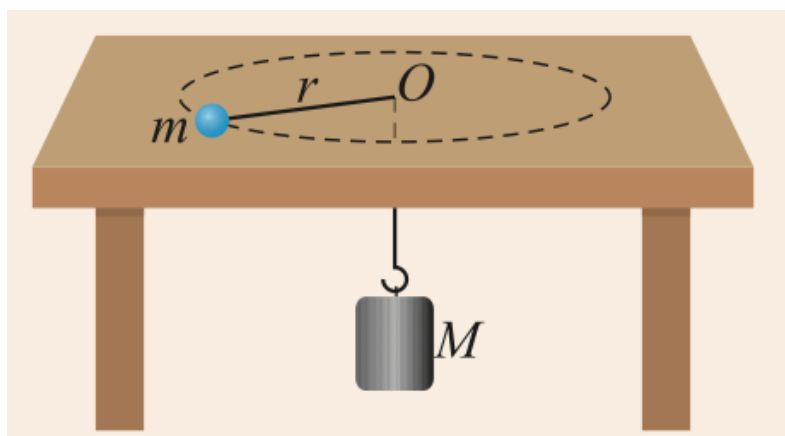
- (a) Calculate the exact value of the normal force.
- (b) Calculate the centripetal force ( $F_c$ ).
- (c) Calculate the centripetal acceleration ( $a_c$ ) of the car.



### Q4

A small blue ball with mass  $m = 5 \text{ kg}$  is on a frictionless horizontal table. It is tied to a string that passes through a hole in the center to a hanging weight with mass  $M = 4 \text{ kg}$ . The blue ball is swinging in a horizontal circle with a radius of  $r = 2 \text{ m}$ .

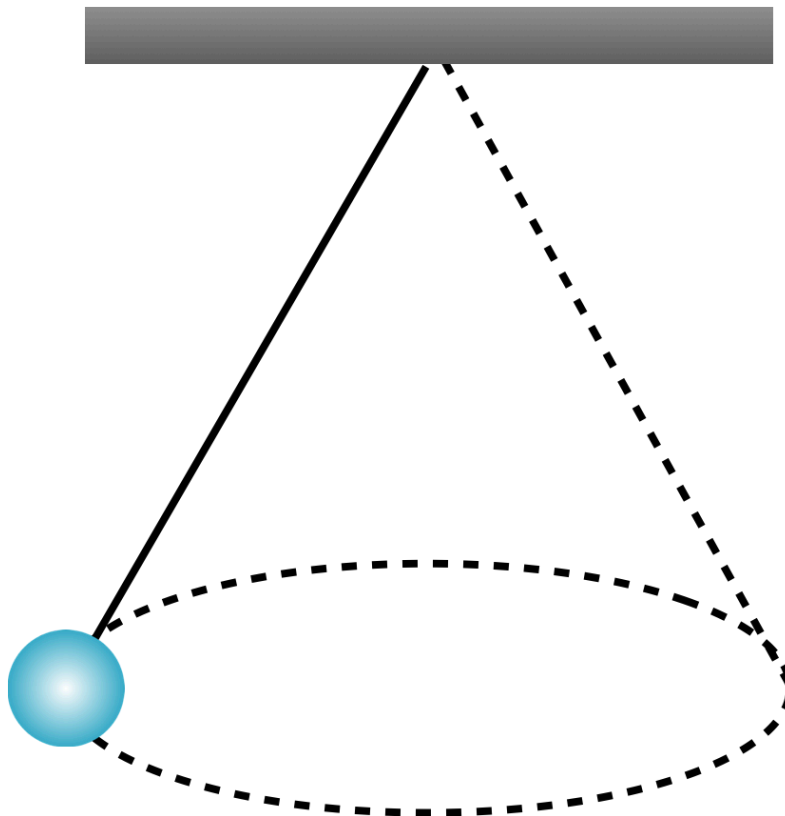
- (a) Calculate the Tension in the string.
- (b) Calculate the centripetal acceleration ( $a_c$ ) of the blue ball.
- (c) Calculate the blue ball's tangential speed.



### Q5

A ball with a mass of  $m = 1.2 \text{ kg}$  is flying in a horizontal circle, suspended by a string of length  $L = 1.5 \text{ m}$ . The string makes an angle of  $\theta = 53^\circ$  with the vertical straight line.

- (a) Calculate the exact value of the tension.
- (b) Calculate the radius  $R$  of the horizontal circle and calculate the centripetal force.
- (c) Calculate the tangential speed of the ball.



### Q6

Analyze the Conical Pendulum further. Imagine you want the same toy plane to fly in a **WIDER circle** (meaning the angle  $\theta$  becomes larger and the plane is higher up).

- (a) Even if the circle is wider, the plane is still making a horizontal circle without falling. Does the vertical upward force ( $T_y$ ) need to change? Explain your reason. (Hint: Look at Gravity).

(b) Because the angle  $\theta$  is larger, the string is more "flat". To keep the SAME vertical force  $T_y$  while pulling at a flatter angle, must the total tension ( $T$ ) in the string increase or decrease? Explain your reason.

(c) To maintain this new, wider circular path, does the plane need to fly faster or slower? Explain your reason.